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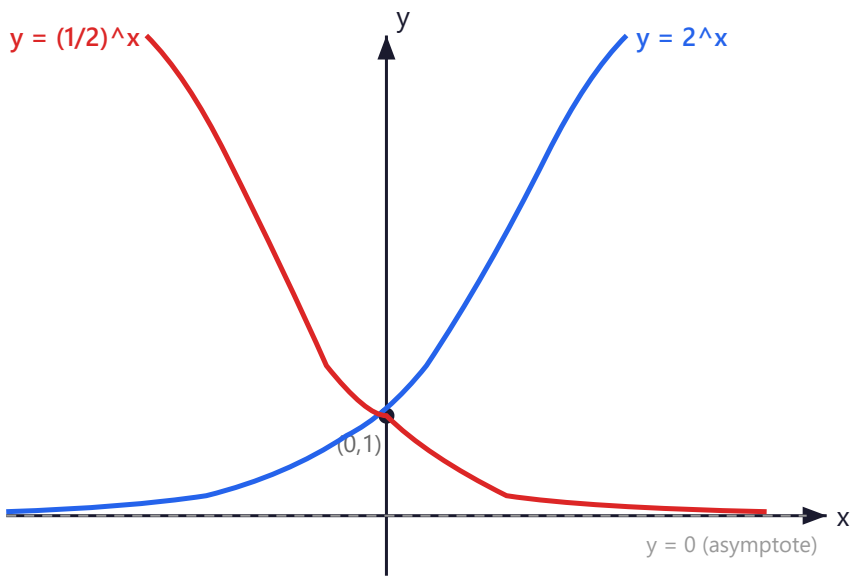
Exponentials & Logarithms

Exponential Functions

Definition: An exponential function has the form $y = a^x$ where $a > 0$ and $a \neq 1$. The base a determines the rate of growth or decay.

Key properties of $y = a^x$:

- The graph always passes through $(0, 1)$ since $a^0 = 1$
- If $a > 1$, the function is increasing (exponential growth)
- If $0 < a < 1$, the function is decreasing (exponential decay)
- The x-axis ($y = 0$) is a horizontal asymptote
- $a^x > 0$ for all values of x



The Natural Exponential Function e^x

Definition: The number $e \approx 2.71828\ldots$ is the unique constant such that the gradient of $y = e^x$ equals e^x at every point. That is, $\frac{d}{dx}(e^x) = e^x$.

The function $y = e^x$ is the most important exponential function in mathematics because of this self-derivative property. Its inverse is the natural logarithm $\ln(x)$.

Logarithms

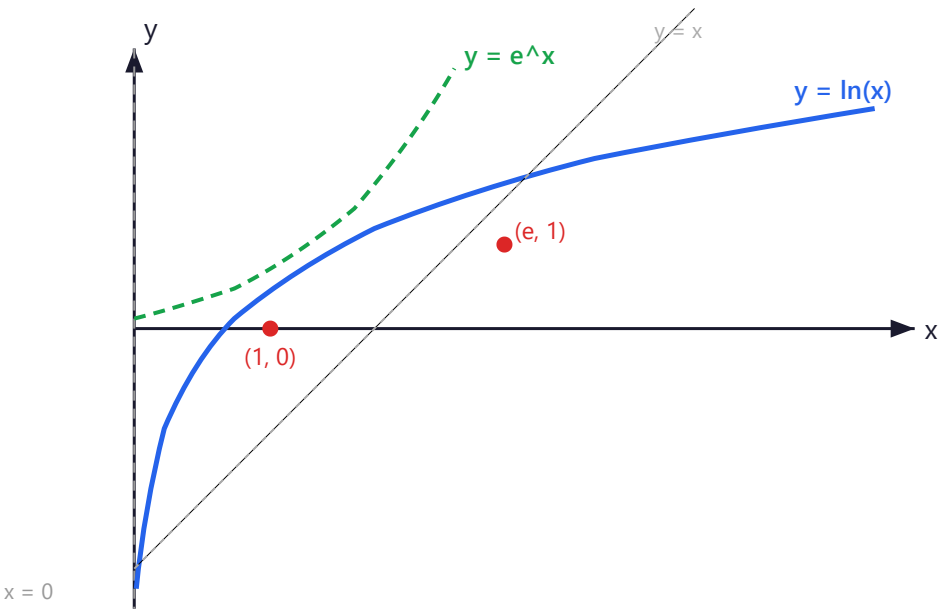
Definition: The logarithm base a of x is the power to which a must be raised to give x:

$$\log_a x = n \iff a^n = x$$

where $a > 0$, $a \neq 1$, and $x > 0$.

Special logarithms:

- $\ln x = \log_e x$ — the natural logarithm (base e)
- $\log x = \log_{10} x$ — common logarithm (base 10)



The graphs of $y = e^x$ and $y = \ln x$ are reflections of each other in the line $y = x$, since they are inverse functions.

Laws of Logarithms

Law	Formula
Product (Multiplication)	$\log_a(xy) = \log_a x + \log_a y$
Quotient (Division)	$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$
Power	$\log_a(x^n) = n\log_a x$
Identity	$\log_a a = 1$
Zero	$\log_a 1 = 0$
Inverse	$a^{\log_a x} = x$ and $\log_a(a^x) = x$

Worked Example: Simplifying with Log Laws

Simplify $\log_2 12 - \log_2 3 + \log_2 8$.

Solution:

Using the quotient and product laws:

$$\log_2 12 - \log_2 3 + \log_2 8 = \log_2\left(\frac{12}{3}\right) + \log_2 8 = \log_2 4 + \log_2 8$$

$$= \log_2(4 \times 8) = \log_2 32 = \log_2 2^5 = 5$$

Exam Tip: When simplifying logarithmic expressions, always look for opportunities to combine terms using log laws before evaluating. Remember that you can only combine logs with the same base.

Solving Logarithmic Equations

To solve equations involving logarithms, use log laws to combine terms, then convert to exponential form.

Worked Example: Combining Logs

Solve $\log_2(x+3) + \log_2(x-1) = 5$.

Solution: Use the product law:

$$\log_2[(x+3)(x-1)] = 5$$

$$(x+3)(x-1) = 2^5 = 32$$

$$x^2 + 2x - 3 = 32 \implies x^2 + 2x - 35 = 0$$

$$(x+7)(x-5) = 0 \implies x = 5 \text{ (reject } x = -7\text{, as log of negative is undefined)}$$

Worked Example: Logarithm Equals Number

Solve $2\ln x - \ln(2x-1) = \ln 3$.

Solution: Use the power and quotient laws:

$$\ln x^2 - \ln(2x-1) = \ln 3 \implies \ln \frac{x^2}{2x-1} = \ln 3$$

$$\frac{x^2}{2x-1} = 3 \implies x^2 = 6x - 3 \implies x^2 - 6x + 3 = 0$$

$$x = \frac{6 \pm \sqrt{24}}{2} = 3 \pm \sqrt{6}$$

Check: for the solution $x = 3 - \sqrt{6} \approx 0.55$, we get $2x - 1 \approx 0.1 > 0$, so both solutions are valid.

Exam Tip: Always check your solutions are valid — a logarithm is only defined for positive arguments. After solving, substitute back and verify that all log arguments are positive. Reject any solution that makes a log argument ≤ 0.

Change of Base Formula

Change of Base:

$$\log_a x = \frac{\log_b x}{\log_b a}$$

In particular, $\log_a x = \frac{\ln x}{\ln a} = \frac{\log x}{\log a}$.

This formula is essential for evaluating logarithms with non-standard bases on a calculator, which typically only has $\ln$ and $\log_{10}$ buttons.

Worked Example: Change of Base

Evaluate $\log_5 20$ to 3 significant figures.

Solution:

$$\log_5 20 = \frac{\ln 20}{\ln 5} = \frac{2.9957...}{1.6094...} = 1.86 \text{ (3 s.f.)}$$

Solving Exponential and Logarithmic Equations

Strategy: Taking Logs of Both Sides

For equations of the form $a^x = b$, take logarithms of both sides:

$$a^x = b \implies x \ln a = \ln b \implies x = \frac{\ln b}{\ln a}$$

Worked Example: Solving $3^{2x+1} = 7^x$

Solution: Take natural logs of both sides:

$$(2x+1)\ln 3 = x \ln 7$$

Expand and collect x terms:

$$2x \ln 3 + \ln 3 = x \ln 7$$

$$x(2\ln 3 - \ln 7) = -\ln 3$$

$$x = \frac{-\ln 3}{2\ln 3 - \ln 7} = \frac{-1.0986...}{2(1.0986...) - 1.9459...} = \frac{-1.0986...}{0.2513...} = -4.37 \text{ (3 s.f.)}$$

Strategy: Substitution for Hidden Quadratics

Worked Example: Hidden Quadratic

Solve $e^{2x} - 5e^x + 6 = 0$.

Solution: Let $u = e^x$, so $e^{2x} = u^2$:

$$u^2 - 5u + 6 = 0$$

$$(u-2)(u-3) = 0$$

So $u = 2$ or $u = 3$, meaning $e^x = 2$ or $e^x = 3$:

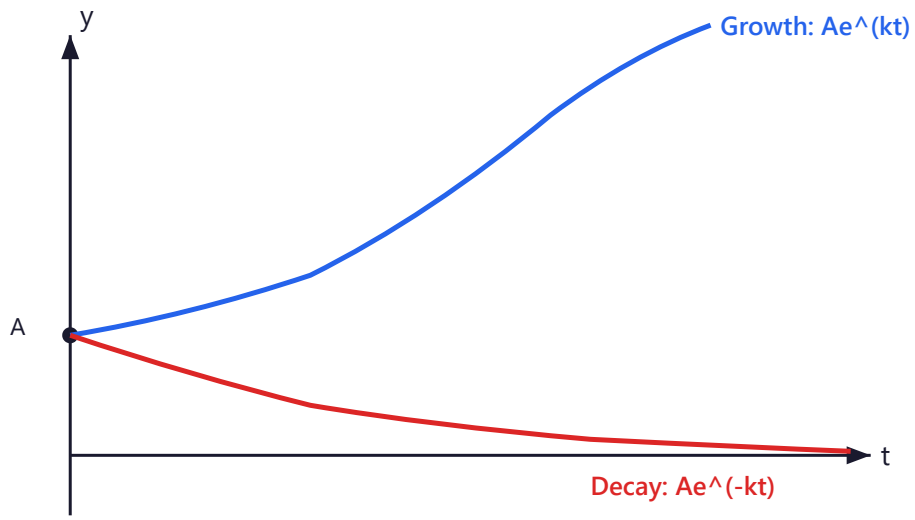
$$x = \ln 2 \approx 0.693 \quad \text{or} \quad x = \ln 3 \approx 1.099$$

Exam Tip: When you see e^{2x}, recognise it as $(e^x)^2$. This substitution technique turns many exponential equations into familiar quadratics.

Exponential Modelling

Many real-world processes follow exponential models:

Exponential Growth: $y = Ae^{kt}$ where $k > 0$, A is the initial value, and t is time.
Exponential Decay: $y = Ae^{-kt}$ where $k > 0$.



Worked Example: Radioactive Decay

A radioactive substance decays according to $M = 200e^{-0.03t}$, where M is the mass in grams and t is time in years.

(a) Find the initial mass.

Solution: When $t = 0$: $M = 200e^0 = 200$ grams.

(b) Find the half-life (time for mass to halve).

Solution: Set $M = 100$:

$$100 = 200e^{-0.03t} \implies e^{-0.03t} = 0.5$$

$$-0.03t = \ln 0.5 \implies t = \frac{\ln 0.5}{-0.03} = \frac{-0.6931...}{-0.03} = 23.1 \text{ years } (3 \text{ s.f.})$$

Linearising Exponential Models

If $y = Ab^x$, take logs of both sides:

$$\log y = \log A + x \log b$$

This is linear in the form $Y = mx + c$, where $Y = \log y$, $m = \log b$, and $c = \log A$.

If $y = ax^n$, take logs:

$$\log y = n \log x + \log a$$

This is linear with gradient n and intercept $\log a$ when plotting $\log y$ against $\log x$.

Worked Example: Linearising Data

Data follows the model $y = 3 \times 2^x$. Describe the straight-line graph obtained by plotting $\log_{10} y$ against x.

Solution:

$$\log y = \log 3 + x \log 2$$

This gives a straight line with gradient $\log 2 \approx 0.301$ and y-intercept $\log 3 \approx 0.477$.

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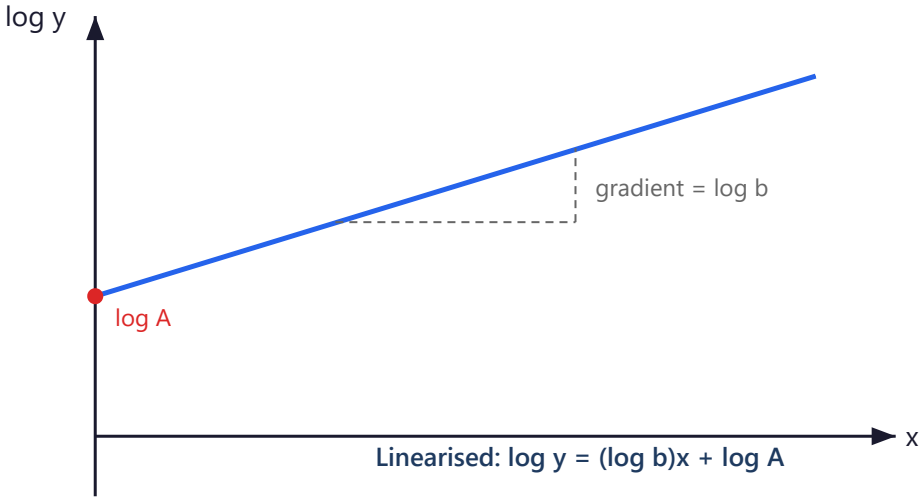

Exam Tip: When a question gives you a straight-line graph of $\log y$ against x (or $\log y$ against $\log x$), read the gradient and intercept carefully to recover the original model parameters. Remember to use $10^{\text{intercept}}$ to find the constant, not just the intercept itself.